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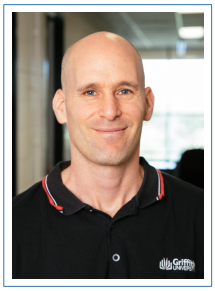

Dr. Ido Bar

Profile Summary

I am a mid-career research leader with extensive expertise in molecular and genomic research in a range of commercially important crops and their pathogens. My work focuses on applying advanced genomics tools to better understand the genes and molecular pathways underlying key bottlenecks in crop production, such as disease resilience and product quality. The knowledge I develop is then used in predictive and pre-breeding applications to support the breeding of sustainable and resilient crops towards better food security in Australia and globally.

Professional Experience

2025–present **Senior Research Fellow**, *School of Environment and Science*, Griffith University, Nathan, QLD, Australia

Lead industry-driven genomic research programs focusing on crop improvement (papaya), disease resilience (chickpea ascochyta, rice bacterial leaf streak), and sustainable aquaculture. My responsibilities include conceptualising and securing industry-funded research grants, designing and overseeing research activities, supervising and mentoring postgraduate students and engaging with industry stakeholders through reporting and delivery of research outputs. Key achievements include securing, leading and delivering the second phase of the GRDC-funded program "Towards effective genetic and sustainable management of Ascochyta blight of chickpea", and co-leading the Hort Innovation program "Building an advanced genomics platform for Australian horticulture" (a collaboration with Prof. Rajeev Varshney at Murdoch University). The outcomes of these projects include the **identification of candidate genes and molecular markers for predictive and pre-breeding for chickpea ascochyta resistance and improved flavour and fruit quality in papaya**.

2021–2024 **Research Fellow**, *Centre for Planetary Health and Food Security*, Griffith University, Nathan, QLD, Australia

Conduct genomic and molecular research on crop improvement (papaya) and host-pathogen interactions (chickpea and lentil ascochyta). My responsibilities included developing versatile bioinformatic analysis pipelines for high-throughput sequencing data in diverse species, primarily crops and their pathogens, and training, mentoring and supervising postgraduate students. Key achievements include delivering research outputs and tools through multiple industry engagement events; developing the Ascochyta Dashboard for data-driven disease management of chickpea ascochyta; and **identifying genes and QTLs associated with resistance to ascochyta blight in lentil and chickpea, and with flavour metabolites in papaya.**

2019–2021 **Lecturer (BioSciences)**, *School of Environment and Science*, Griffith University, Nathan, QLD, Australia

Designed and delivered undergraduate and postgraduate courses in Molecular Biology, Applied Bioinformatics, and Scientific Data Analysis. Earned the **Head of School Teaching Commendation** for outstanding student evaluations. Pioneered the use of cloud-based platforms (Binder, Nectar) to enhance online learning and promoted the adoption of FAIR and open data practices across the university.

2016–2018 **Postdoctoral Research Fellow**, *Environmental Futures Research Institute*, Griffith University, Nathan, QLD, Australia

Conducted foundational bioinformatics research on crop and pathogen species. Specialised in analysing high-throughput sequencing data (WGS, RNA-Seq, GBS) to investigate disease resistance, identify genetic markers, and map population structures. Developed automated data analysis pipelines using R, Python and Bash in HPC environments and created novel bioinformatic tools (nf-taxblast and Shinotate) to accelerate genome and transcriptome annotation. Provided bioinformatics and data analysis training and mentorship to students and staff.

Education

2012–2016 **PhD – Molecular Biology and Genomics**, *Faculty of Science, Health, Education and Engineering, Genecology Research Centre, University of the Sunshine Coast, QLD, Australia*

2009–2011 **MSc Agri – Animal Sciences**, *Faculty of Agricultural, Food and Environmental Quality Sciences, Hebrew University of Jerusalem*, Rehovot, Israel

Final grade *92.5/100 (Cum Laude)*

2003–2007 **BSc – Marine Biology and Biotechnology**, *Faculty of Life Sciences, Ben Gurion University of the Negev*, Beer-Sheva, Israel

Degree average *90/100*

Training and Certifications

2021 New Leader course (Australian Institute of Management)

2018 Software Carpentry Instructor Training (QFAB workshop)

2016 Hypothesis Testing Using R (QFAB workshop)

2014 Data Analysis for Genomics (HarvardX course PH525x)

2014 Extended RNA-Seq Analysis (QFAB workshop)

2011 Genomics in Biodiversity (MARBIGEN workshop)

Research Outputs

I have published a total of 2 book chapters, 36 peer-reviewed journal articles and several preprints, code packages and datasets, with >850 citations and an h-index of 17. These publications were mostly published in Q1 journals. The following key publications highlight my expertise in applying genomics in crop research. For a complete publication record, see my Google Scholar profile.

Books Chapters

2020 Dorin Gupta, Rama Harinath Dadu, Prabhakaran Sambasivam, **Ido Bar**, Mahsa Azad, Navya Beera, Rebecca Ford, and Sajitha Biju. "Conventional and Biotechnological Approaches for Targeted Trait Improvement in Lentil". In: *Accelerated Plant Breeding, Volume 3: Food Legumes*. Ed. by Satbir Singh Gosal and Shabir Hussain Wani. Cham: Springer International Publishing, 2020, pp. 67–107

2019 Dorin Gupta, Rama Harinath Reddy Dadu, Prabhakaran Sambasivam, **Ido Bar**, Mohar Singh, Navya Beera, and Sajitha Biju. "Toward Climate-Resilient Lentils: Challenges and Opportunities". In: *Genomic Designing of Climate-Smart Pulse Crops*. Ed. by Chittaranjan Kole. Cham: Springer International Publishing, 2019, pp. 165–234

Journal Publications

2025 Mahmuda Binte Monsur, **Ido Bar**, Jonathan Wanderley Lawley, and Rebecca Ford. "Effector Molecules and Pathogenicity-Associated Gene Expression in *Ascochyta rabiei*". In: *Fungal Biology* (Oct. 5, 2025), p. 101668

2025 Moe Moe Kyi Win, Wanchana Aesomnuk, Reajina Dumhai, Siriphat Ruengphayak, Julie E. Gray, **Ido Bar**, Vinitchan Ruanjaichon, Samart Wanchana, and Siwaret Arikit. "Transcriptomic Analysis of Rice Cultivars with Distinct Resistance Mechanisms to *Xanthomonas oryzae* Pv. Oryzicola Reveals Novel Components and Candidate Genes Associated with Bacterial Leaf Streak". In: *Frontiers in Plant Science* 16 (Sept. 9, 2025)

2025 Hayley Wilson, **Ido Bar**, Kristy Hobson, Niloofar Vaghefi, Toni Petronitis, and Rebecca Ford. "Integrated Disease Management, Adaptation and Genomics of Fungal Plant Pathogens in Cropping Systems". In: *Plant Pathology* (May 12, 2025)

2024 Niloofar Vaghefi, **Ido Bar**, Jonathan Wanderley Lawley, Prabhakaran Thanjavur Sambasivam, Melody Christie, and Rebecca Ford. "Population-Level Whole-Genome Sequencing of *Ascochyta rabiei* Identifies Genomic Loci Associated with Isolate Aggressiveness". In: *Microbial Genomics* 10.11 (Nov. 22, 2024), p. 001326

2021 Rama Harinath Reddy Dadu, **Ido Bar**, Rebecca Ford, Prabhakaran Sambasivam, Janine Croser, Federico Ribalta, Sukhjiwan Kaur, Shimna Sudheesh, and Dorin Gupta. "*Lens orientalis* Contributes Quantitative Trait Loci and Candidate Genes Associated With Ascochyta Blight Resistance in Lentil". In: *Frontiers in Plant Science* 12 (2021), p. 1784

2019 Ziwei Zhou, **Ido Bar**, Prabhakaran Thanjavur Sambasivam, and Rebecca Ford. "Determination of the Key Resistance Gene Analogs Involved in *Ascochyta rabiei* Recognition in Chickpea". In: *Frontiers in Plant Science* 10 (2019)

2018 Mahsa Khorramdelazad, **Ido Bar**, Paul Whatmore, Gabrielle Smetham, Vijay Bhaaskaria, Yuedong Yang, Shahla Hosseini Bai, Nitin Mantri, Yaoqi Zhou, and Rebecca Ford. "Transcriptome profiling of lentil (*Lens culinaris*) through the first 24 hours of *Ascochyta lentis* infection reveals key defence response genes". In: *BMC Genomics* 19 (2018), p. 108

Multi-Omic Research in Papaya

2026 Joshua Mark Hillis Lomax, Rebecca Ford, Joana Pico, Heather Smyth, and **Ido Bar**. "Metabolomic Modelling of Sensory Characteristics and Consumer Liking in Papaya Fruit". In: *Food Chemistry* 508 (Apr. 15, 2026), p. 148323

2025 Rhys Glen Richard Copeland, **Ido Bar**, Shri Hari Prasad, Reyazul Rouf Mir, Vanika Garg, Robert Henry, Ray Ming, and Rajeev Kumar Varshney. "Harnessing Novel Genomic Resources and Emerging Breeding Tools to Fast-Track Genetic Improvement in Papaya". In: *Horticultural Plant Journal* (July 23, 2025)

2024 Joshua Lomax, Rebecca Ford, and **Ido Bar**. "Multi-Omic Applications for Understanding and Enhancing Tropical Fruit Flavour". In: *Plant Molecular Biology* 114.4 (Aug. 2024), p. 83

2024 Ziwei Zhou, Chutchamas Kanchana-udomkan, Rebecca Ford, and **Ido Bar**. "Identification and Validation of Key Genes Related to Preferred Flavour Profiles in Australian Commercial Papaya (*Carica papaya* L.)". In: *International Journal of Molecular Sciences* 25.5 (5 Mar. 6, 2024), p. 3046

2019 Usana Nantawan, Chutchamas Kanchana-udomkan, **Ido Bar**, and Rebecca Ford. "Linkage Mapping and Quantitative Trait Loci Analysis of Sweetness and Other Fruit Quality Traits in Papaya". In: *BMC Plant Biology* 19.1 (2019), p. 449

Conference Presentations

Selected presentations from the past 5 years:

IFLRC 2025 *Invited Talk*: Clone Wars: The story of *Ascochyta rabiei* in Australia. International Food Legume Research Conference 2025, Perth, WA, Australia.

TropAg 2025 *Oral Presentation*: Genomics of fruit flavour in Australian Papaya towards targeted breeding of elite varieties. TropAg Conference 2025, Brisbane, QLD, Australia.

NPP 2024 *Oral Presentation*: DNA methylation and its role in the pathogenicity of *Ascochyta rabiei* isolates. National Pulse Pathology meeting 2024, Brisbane, QLD, Australia.

PAG 2023 *Oral Presentation*: Developing genomic resources for Australian Papaya towards targeted breeding of elite varieties. Plant and Animal Genome Conference 2023, Perth, WA, Australia.

ICPP 2023 *Oral Presentation*: Comparative genomics reveals key genetic polymorphisms associated with increased aggressiveness in Australian *Ascochyta rabiei*. International Congress of Plant Pathology 2023, Lyon, France

ICPP 2023 *Poster*: Asco dashboard — an online interactive tool for exploring and investigating *Ascochyta rabiei* populations in Australia. International Congress of Plant Pathology 2023, Lyon, France

APBC 2022 *Oral Presentation*: Genetics of Fruit Sensory Preferences. Australasian Plant Breeding Conference 2022, Gold Coast, QLD, Australia

Industry Engagement & Impact

Research Grants

2024–2027 Grain Research and Development Corporation – Towards effective genetic and sustainable management of Ascochyta blight of chickpea, Phase II (PI; $570k; GRI2407-001RTX)

2020–2024 Grain Research and Development Corporation – Towards effective genetic and sustainable management of Ascochyta blight of chickpea (Co-PI; $1.2m; GRI2007-001RTX)

2024–2027 Advance Queensland Industry Research Fellowship – Invigorating the coffee industry in Tropical North Queensland, AQIRF007-2023RD6 (Co-investigator; $646k)

2023–2027 Hort Innovation – Building an advanced genomics platform for Australian horticulture (Co-PI; $802k; AS21006)

2023–2024 The Crawford Fund International Engagement Award – Identification of novel rice resistance genes to bacterial leaf streak disease through transcriptomic analysis (PI; $12k)

2022–2023 The Crawford Fund International Engagement Award – Genome-wide association study for anthracnose resistance and fruit quality traits in Thai chilli germplasm (PI; $12k)

2022–2025 NSW Department of Primary Industries – Investigation into the evolution of increased *Ascochyta rabiei* aggressiveness against chickpea (Co-PI; $70k)

2021–2026 Hort Innovation – Genetics of fruit sensory preferences (PI; $466k; AS19003)

Industry & Media Engagement

Apr 2025 Cracking The Papaya Flavour Code (Papaya Press)

Jul 2024 Practical management of Ascochyta in chickpeas (GRDC Research Updates webinar and in person at the Gindi-in-July and Walgett Northern Region Updates)

Aug 2022 Dashboard tracks rising Ascochyta aggression (GRDC GroundCover)

Aug 2022 Fighting back against the chickpea scourge (GRDC GroundCover)

Jan 2022 Growing Fruit of the Future (The Rural News at 2GB Radio)

Nov 2021 Scientists imagine a future where consumers always like fruit (Hort Innovation media release)

Sep 2021 Chickpea Challenges: The highs and lows of growing chickpeas (ABC landline)

Professional Skills

Wet Lab

Mol. Biology Skilled in a wide range of molecular biology experimental approaches, including nucleic acid extraction, purification and quantification (with Nanodrop, Qubit and qPCR), PCR primer design, amplification and optimisation, gene *in-situ* hybridisation, ELISA, plasmid design and cloning

Genomics Designing and executing experiments for High-throughput Sequencing (HTS) analyses, including RNA-Seq, Whole-Genome-Sequencing, Genotyping-by-Sequencing and methylation, using Illumina, MGI, DArT, Nanopore and PacBio sequencing platforms

Computational Biology for Large-Scale Datasets

NGS Analysis	Experienced in developing and running complex NGS/HTS data analysis pipelines including QC, transcriptome, genome and pan-genome assembly and annotation, variant detection, phylogeny and population genetics analysis, QTL mapping, GWAS, differential expression and methylation analyses.
Genetics Analysis	Sequence analysis (BioEdit, MEGA, Geneious), microsatellite allele calling (GeneMarker), GenAlEx, STRUCTURE, poppr, adegenet, SNPRelate, R/qtl, and many more

Computer Skills

Programming	Highly experienced in developing data processing scripts in R, Python, Bash (sed, awk, GNU-Parallel, etc.), using version control (GitHub profile). Experienced in the design and implementation of relational databases (MariaDB, MySQL, SQLite)
Data Analysis	High-level data analysis, including the processing and analysis of large genomic datasets in HPC environments (PBS, SLURM) using advanced workflow managers (Nextflow) and containerised software (Docker and Singularity), followed by application of appropriate statistical methods and tests, data summary and visualisation using R (primarily) and MS-Excel
Reporting	Compile professional scientific documents and reports using MS-Office, LaTeX and bioinformatics workflows documentation using Rmarkdown (see example); develop and deploy R-Shiny web applications and dashboards (Ascochyta Dashboard)
OS	Windows, Linux

Interests and Voluntary Activities

Committees

2025	Member of the scientific committee of the 2025 International Food Legume Society conference
2023–2025	Academic advisor to Griffith eResearch regarding research-focused computing infrastructure and digital solutions purchases and usage, Griffith University
2021–2024	Member of the management committee and theme leader for "Sustainable Agriculture" at the Centre for Planetary Health and Food Security, Griffith University
2022	Member of the organising committee of the 2022 National Chickpea Ascochyta Blight Workshop
2019–2021	Member of the Early Career Researchers committee, Griffith Sciences

Community

2018–Present	Provide bioinformatics, digital literacy and eResearch support and training for students and researchers through Software Carpentry, HackyHour and ResBaz meetings and workshops
2015–Present	Manuscript reviewer and guest editor for Q1 journals such as BMC Genomics, Frontiers, PLoS ONE, Scientific Reports, Genes and others (see details in my WoS profile)

Hobbies

Sea and whitewater kayaking, fishing, camping and basically any outdoor activity